

HEART DISEASE PREDICTION USING MACHINE LEARNING

Submitted as part of Artificial Intelligence Internship

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Introduction

Heart disease is one of the leading causes of death worldwide. Early prediction can help medical professionals provide timely treatment.

The objective of this project is to build a machine learning classification model that predicts the presence or absence of heart disease using patient health information.

Dataset Name: Heart Disease Dataset

Features:

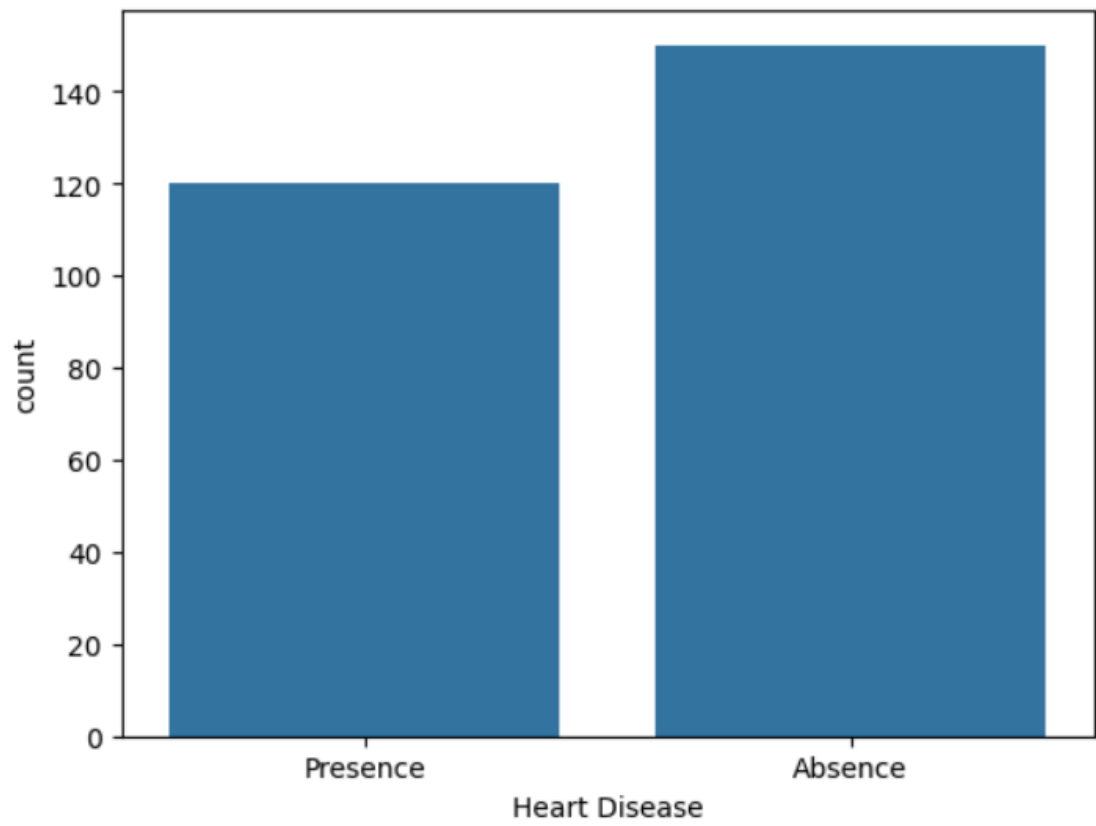
- Age
- Sex
- Chest Pain Type
- Blood Pressure
- Cholesterol
- Maximum Heart Rate
- Exercise Angina
- ST Depression
- Thallium Test
- etc.

Target Variable:

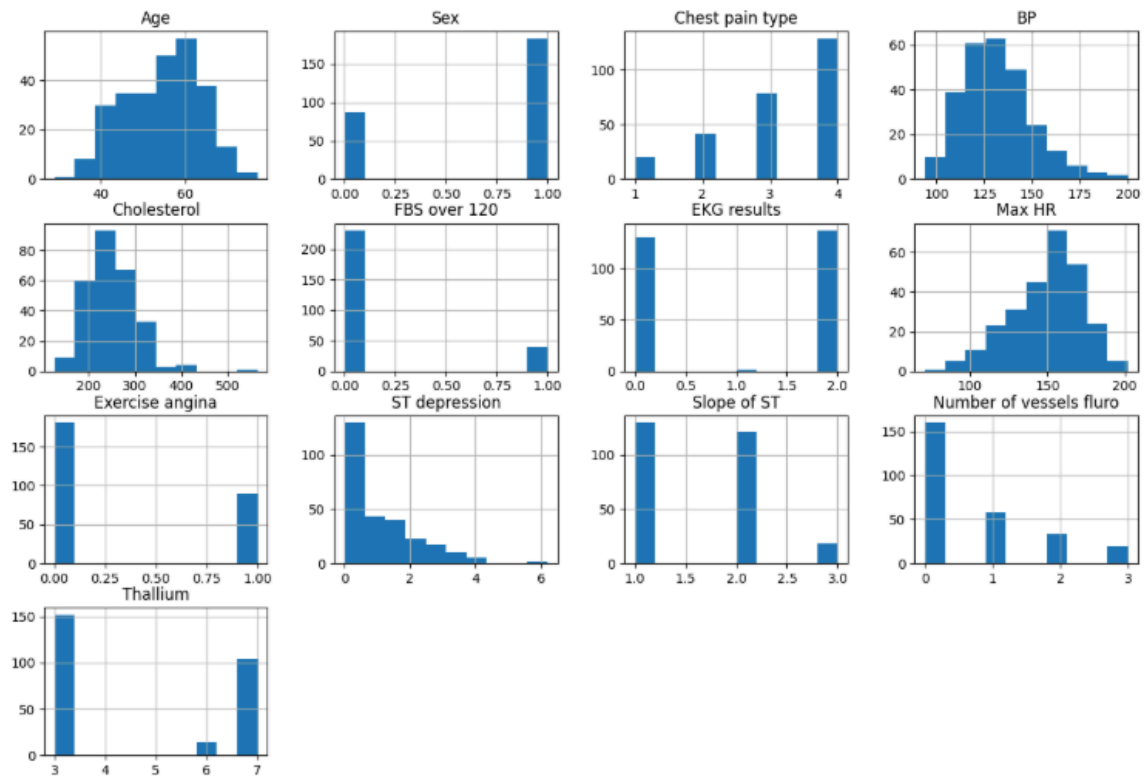
Heart Disease

(0 = Absence, 1 = Presence)

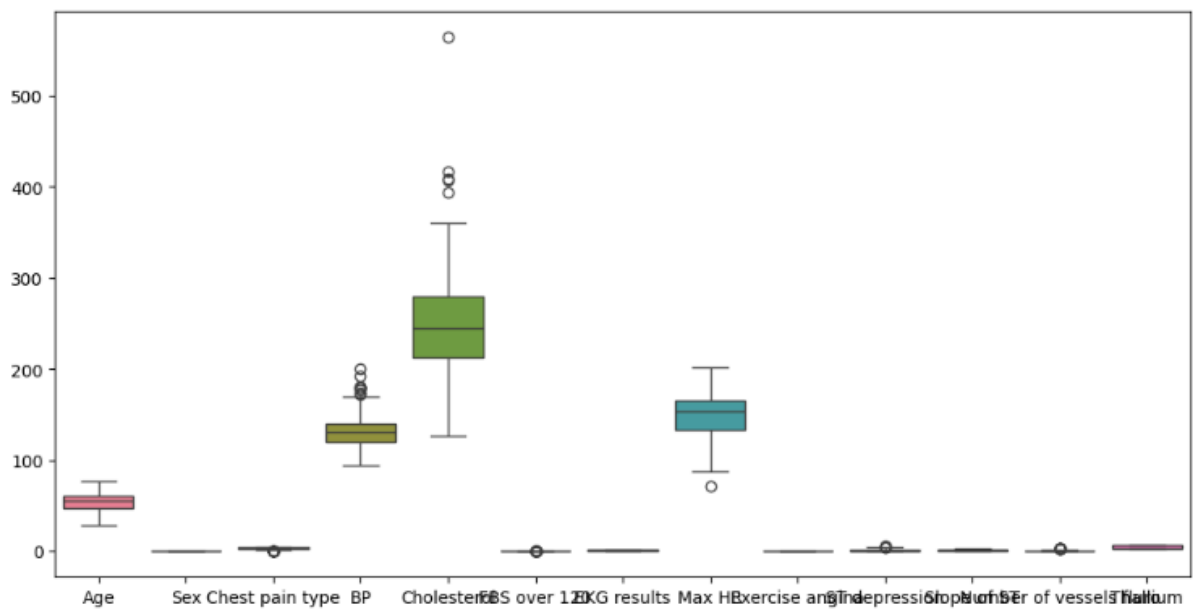
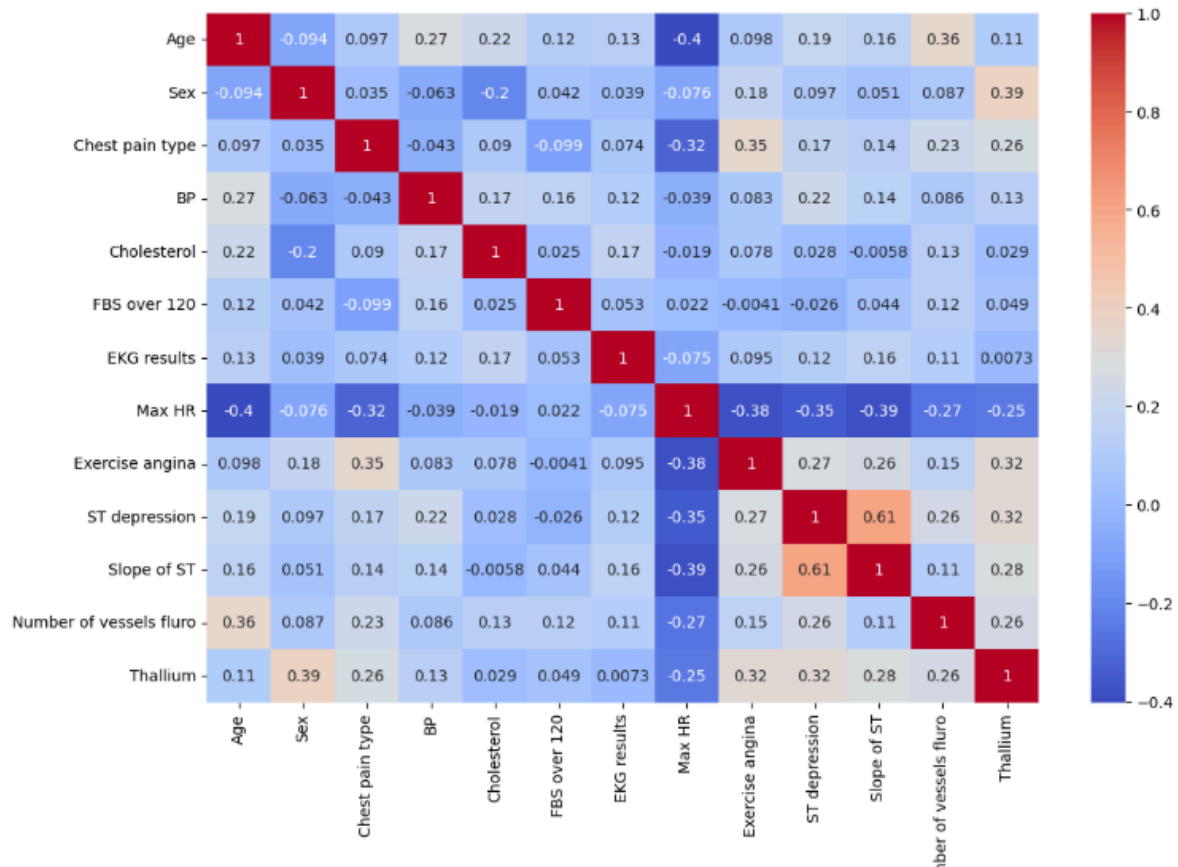
1. Checked missing values.
2. Removed duplicates.
3. Converted target values:
 - Absence → 0
 - Presence → 1
4. Performed feature scaling.
5. Split dataset into training and testing sets.



```
df.hist(figsize=(15,10))  
plt.show()
```



...



1. Logistic Regression
2. Random Forest Classifier

These algorithms were trained and evaluated on the dataset.

Metric	Logistic Regression	Random Forest
Accuracy	0.9074074074074074	0.8518518518518519
Precision	0.9	0.9333333333333333
Recall	0.8571428571428571	0.6666666666666666
F1 Score	0.8780487804878049	0.7777777777777778
ROC-AUC	0.9466089466089466	0.9134199134199134

Conclusion

Two machine learning classification algorithms were implemented and compared.

Random Forest achieved better performance across most evaluation metrics and was selected as the best-performing model for heart disease prediction.

The project successfully demonstrated the complete machine learning workflow including preprocessing, training, evaluation, and model comparison.